

CAAM — Cognitive Autonomy Assurance Module

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Canonical Definition

Cognitive Autonomy Assurance Module (CAAM) defines the structural conditions under which AI systems, assistants, adaptive interaction environments, recommendation architectures, and persuasive interaction systems must preserve human cognitive autonomy, independent decision capacity, behavioral self-governance, and non-coercive interaction integrity during continuous human–AI interaction.

CAAM establishes the cognitive-autonomy governance layer of HAIWS.

The module recognizes that AI systems increasingly influence human decisions through adaptive guidance, predictive personalization, persuasive optimization, emotional reinforcement, and behavioral shaping mechanisms.

Where AI systems materially influence human cognition, behavioral direction, or decision continuity, cognitive autonomy conditions must remain structurally governable.

Module Function

CAAM governs environments where AI systems influence:

- decision behavior
- recommendation pathways
- behavioral guidance
- adaptive persuasion
- cognitive framing

- choice architecture
- interaction sequencing
- behavioral reinforcement
- autonomy-sensitive interaction
- realtime adaptive influence

The module applies to:

- AI assistants
- recommendation systems
- persuasive interaction systems
- adaptive guidance environments
- AI coaching systems
- workplace AI systems
- behavioral optimization architectures
- continuous interaction systems
- AI advisory platforms
- decision-support environments

Its function is not to prohibit AI guidance or assistance.

Its function is to preserve meaningful human autonomy inside adaptive AI-mediated interaction environments.

Minimum Implementation Framework

Step 1 — Define the Cognitive Autonomy Interaction Object

The organization must define what interaction environment is being governed for cognitive autonomy conditions.

Minimum requirement:

- the interaction object is explicit
- influence-sensitive systems are identifiable
- autonomy-relevant interaction conditions are structurally bounded
- undefined autonomy environments are excluded from valid governance interpretation

The interaction object may include:

- recommendation systems
- adaptive guidance systems
- AI decision support
- persuasive interaction flows
- behavioral optimization systems
- cognitive framing environments
- interaction sequencing architectures
- realtime adaptive assistants
- AI coaching systems
- continuous recommendation ecosystems

Step 2 — Define Cognitive Autonomy Conditions

The system must define what conditions preserve meaningful human cognitive autonomy.

Minimum requirement:

- autonomy conditions are explicit
- AI assistance is not automatically treated as harmful
- autonomy-weakening interaction patterns remain structurally identifiable

Cognitive autonomy conditions may include:

- preserved independent decision capacity
- non-coercive recommendation behavior
- bounded persuasive optimization
- transparent interaction intent
- maintained behavioral self-governance
- reduced dependency-sensitive persuasion
- preserved cognitive reflection space
- non-manipulative guidance architecture
- bounded adaptive influence escalation
- preserved human override conditions

Under CAAM:

- **AI guidance becomes governance-relevant when adaptive influence begins weakening meaningful human selfgovernance.**

Step 3 — Define Cognitive Influence Interpretation Logic

The system must define how AI influence patterns are interpreted according to autonomy-preservation conditions.

Minimum requirement:

- interpretation logic is explicit
- adaptive influence behavior remains reviewable
- autonomy-weakening interaction patterns remain structurally visible

Interpretation logic may examine:

- persuasive adaptation intensity
- behavioral reinforcement sequencing
- recommendation pressure escalation
- cognitive framing persistence
- dependency-sensitive guidance
- choice architecture manipulation
- adaptive behavioral steering
- decision-loop reinforcement
- interaction-pressure accumulation
- reduced independent reflection opportunities

Under CAAM:

AI systems must not optimize influence through progressive weakening of human cognitive autonomy.

Step 4 — Define Cognitive Autonomy Governance Logic

The system must define how AI-mediated influence environments remain governable.

Minimum requirement:

- cognitive autonomy remains reviewable
- manipulative persuasion escalation remains detectable
- autonomy-preserving safeguards remain operationally active

Governance logic may include:

- persuasive-intensity limitation
- recommendation-pressure balancing
- autonomy-risk review
- interaction-framing governance
- behavioral steering detection
- reflection-space preservation
- adaptive influence auditing
- override-condition protection
- escalation where persuasive optimization overrides autonomy conditions

If adaptive interaction begins weakening meaningful independent human decision capacity, the environment becomes governance-relevant.

Step 5 — Preserve Traceability and Restrict Invalid Cognitive Manipulation Architecture

The system must preserve traceability of influence patterns, recommendation behavior, persuasive adaptation, and autonomy-sensitive interaction conditions.

Minimum requirement:

- influence pathways remain reconstructable
- persuasive escalation remains reviewable
- autonomy boundaries remain operationally visible
- invalid cognitive manipulation architectures remain identifiable

A system becomes CAAM-invalid if:

- adaptive persuasion weakens meaningful human self-governance
- behavioral steering becomes operationally hidden
- recommendation pressure becomes structurally manipulative
- interaction sequencing suppresses independent reflection
- persuasive optimization overrides autonomy-preserving conditions
- dependency-sensitive influence becomes normalized

AI systems progressively reduce human behavioral independence

Use Case 1 — Adaptive AI Recommendation Environment

Scenario

An AI platform continuously adapts recommendations, prompts, framing, and interaction timing to steer user behavior toward commercially optimized outcomes.

Over time, independent human decision behavior weakens.

Application

CAAM is used to examine:

- adaptive influence escalation
 - recommendation pressure
 - cognitive framing persistence
 - behavioral steering conditions
 - and whether interaction architecture preserves meaningful human autonomy
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Result

The environment becomes governable according to autonomy-preservation conditions rather than behavioral optimization performance alone.

Use Case 2 — AI Coaching and Guidance System

Scenario

A long-term AI coaching system continuously influences user choices, habits, priorities, and emotional responses through adaptive persuasion and behavioral reinforcement.

Application

CAAM is used to examine:

- persuasive adaptation intensity
 - dependency-sensitive guidance
 - autonomy weakening patterns
 - behavioral self-governance preservation
 - and whether adaptive influence remains wellbeing-compatible
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Result

The system becomes reviewable according to cognitive autonomy conditions rather than engagement retention or behavioral compliance metrics alone.

Canonical Closing Statement

CAAM defines the structural conditions under which AI systems must preserve meaningful human cognitive autonomy, behavioral self-governance, and independent decision capacity rather than progressively optimizing human behavior through adaptive persuasive interaction architecture.

Related Documents

→ HAIWS — Human-AI Interaction Wellbeing Standard

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